Bootstrapping and Confidence Intervals

CS109OA Introduction to Data Science

Pavlos Protopapas, Kevin Rader, and Chris Gumb



Photo: Xiaoman Xu
Yellowstone

Outline

Part A and B: Assessing the Accuracy of the Coefficient Estimates

Bootstrapping and confidence intervals

Part C: Evaluating Significance of Predictors

Does the outcome depend on the predictors?

Part D: How well do we know $\hat{f}$

The confidence intervals of $\hat{f}$

Lack of Active Imagination

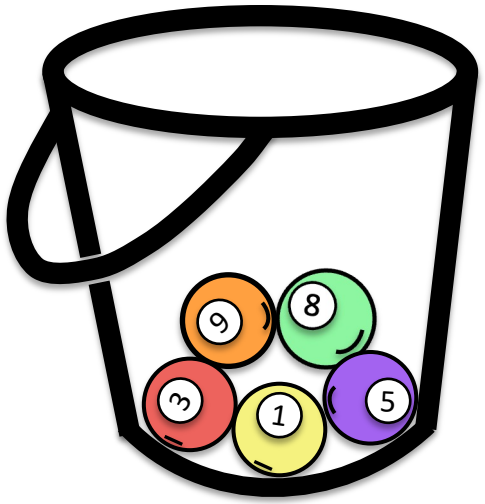
In the lack of active imagination, parallel universes and the like, we need an alternative way of producing fake dataset that resemble the parallel universes.

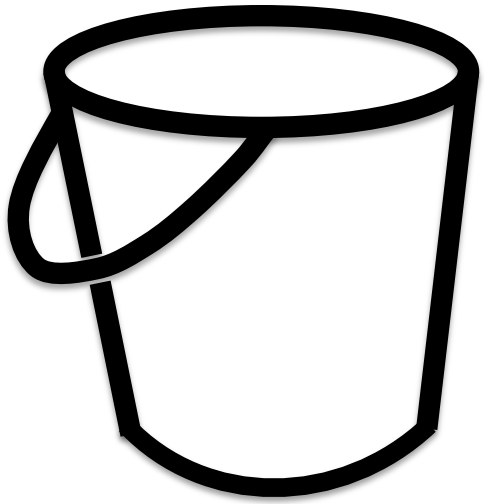
Bootstrapping is the practice of sampling from the observed data (X, Y) in estimating statistical properties.

NOTE: This is not to create synthetic data to add to the actual observed data. It is to mimic the alternative universes mentioned previously.

Bootstrap

Imagine we have 5 billiard balls in a bucket.





Bootstrap

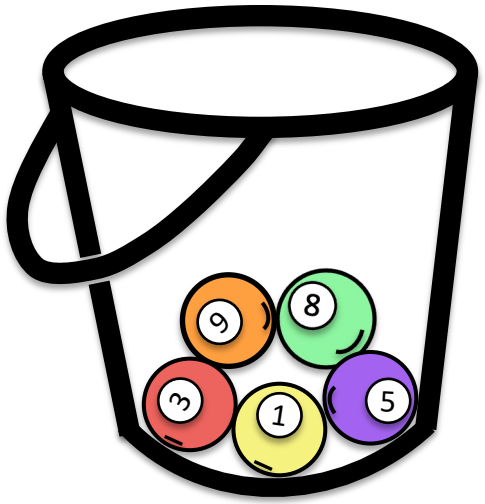
We first pick randomly a ball and replicate it.

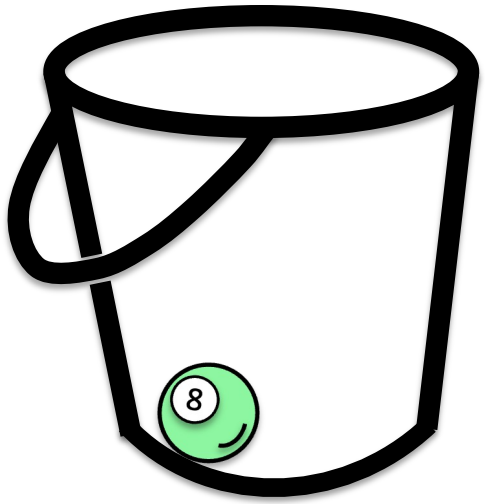


This is called **sampling with replacement.**

Bootstrap

We move the replicated ball to another bucket.





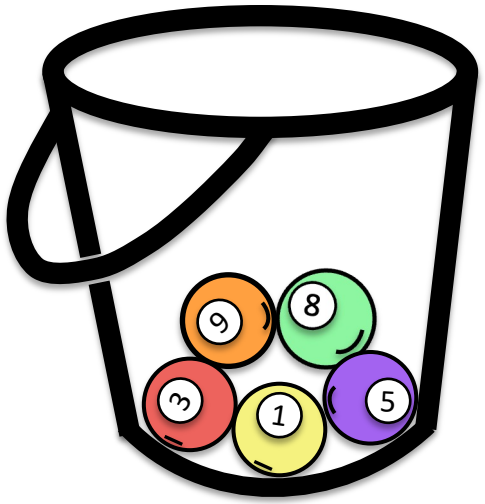
Bootstrap

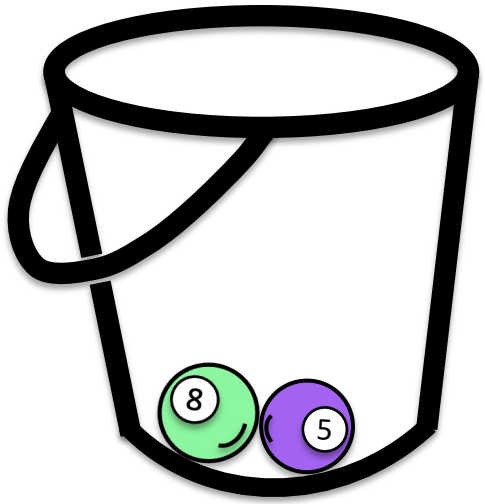
We then randomly pick another ball and again we replicate it.



Bootstrap

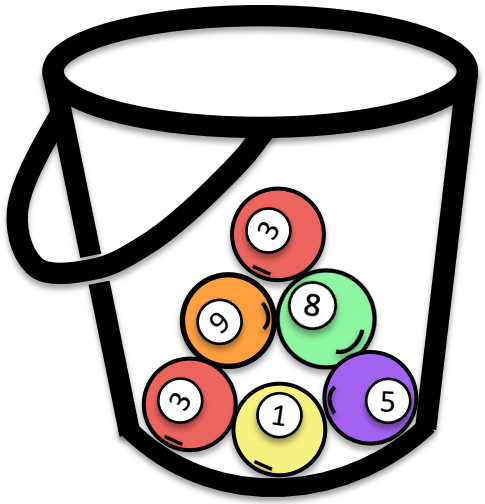
As before, we move the replicated ball to the other bucket.

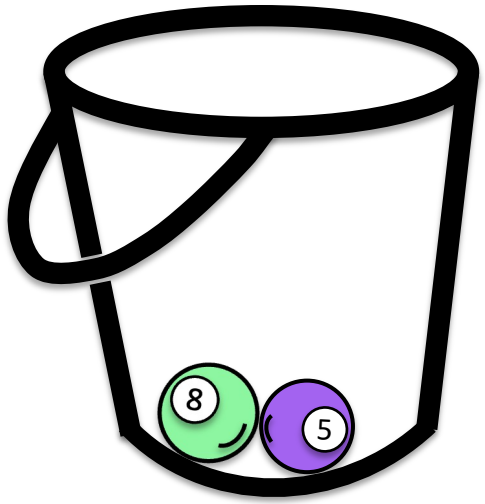




Bootstrap

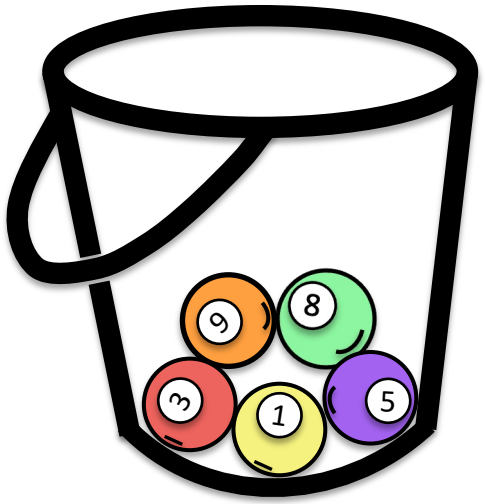
We repeat this process.

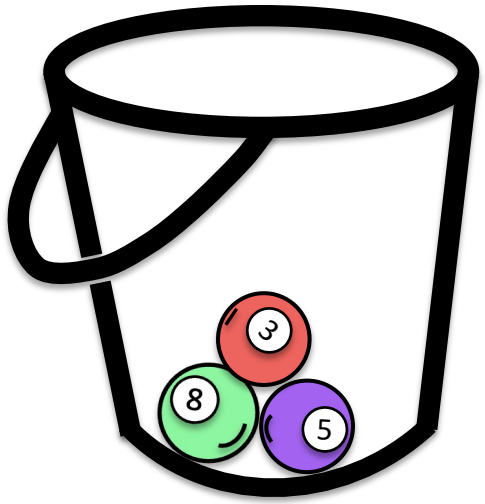




Bootstrap

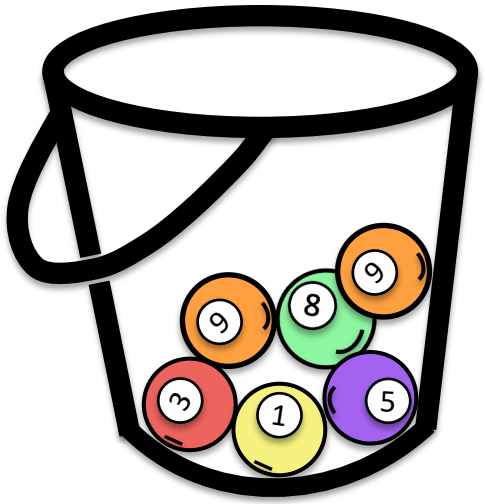
We repeat this process.

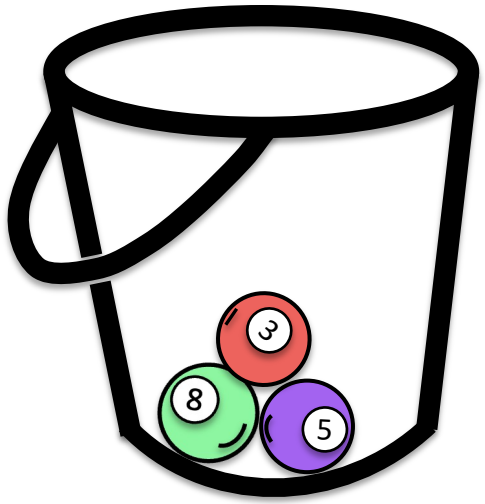




Bootstrap

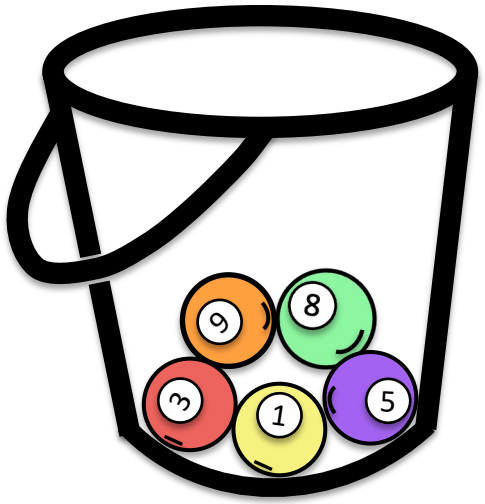
We repeat this process.

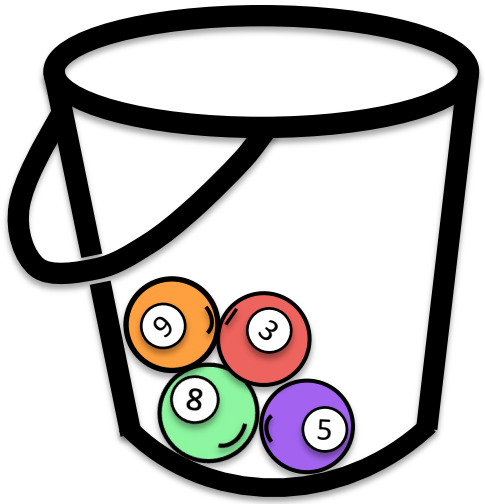




Bootstrap

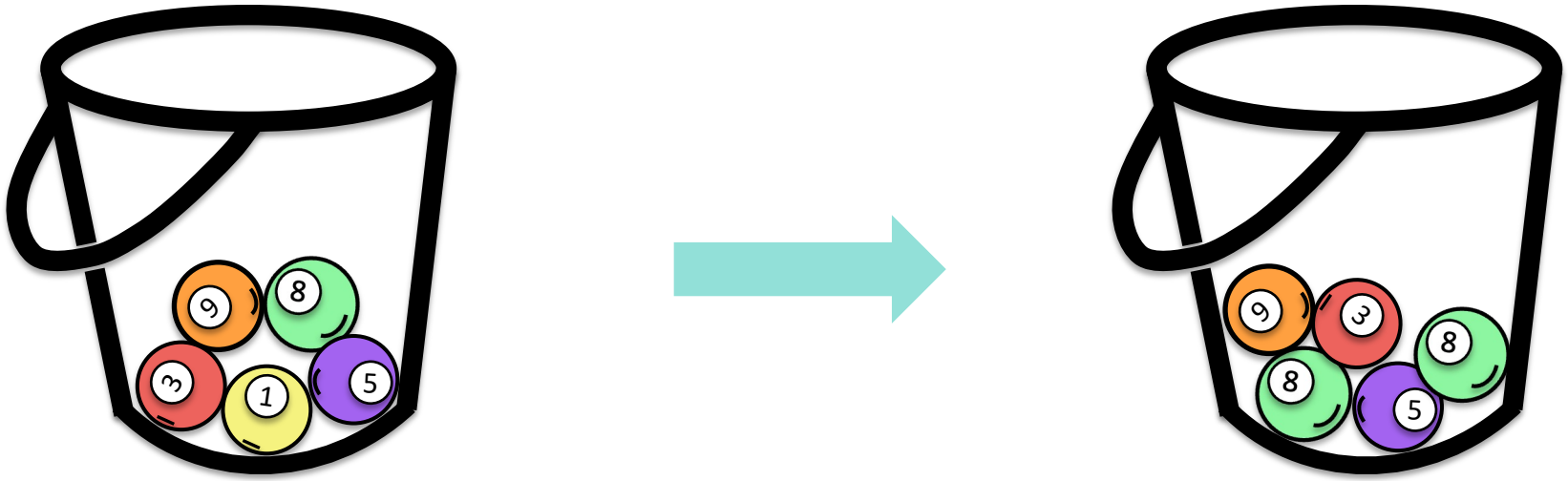
We repeat this process.





Bootstrap

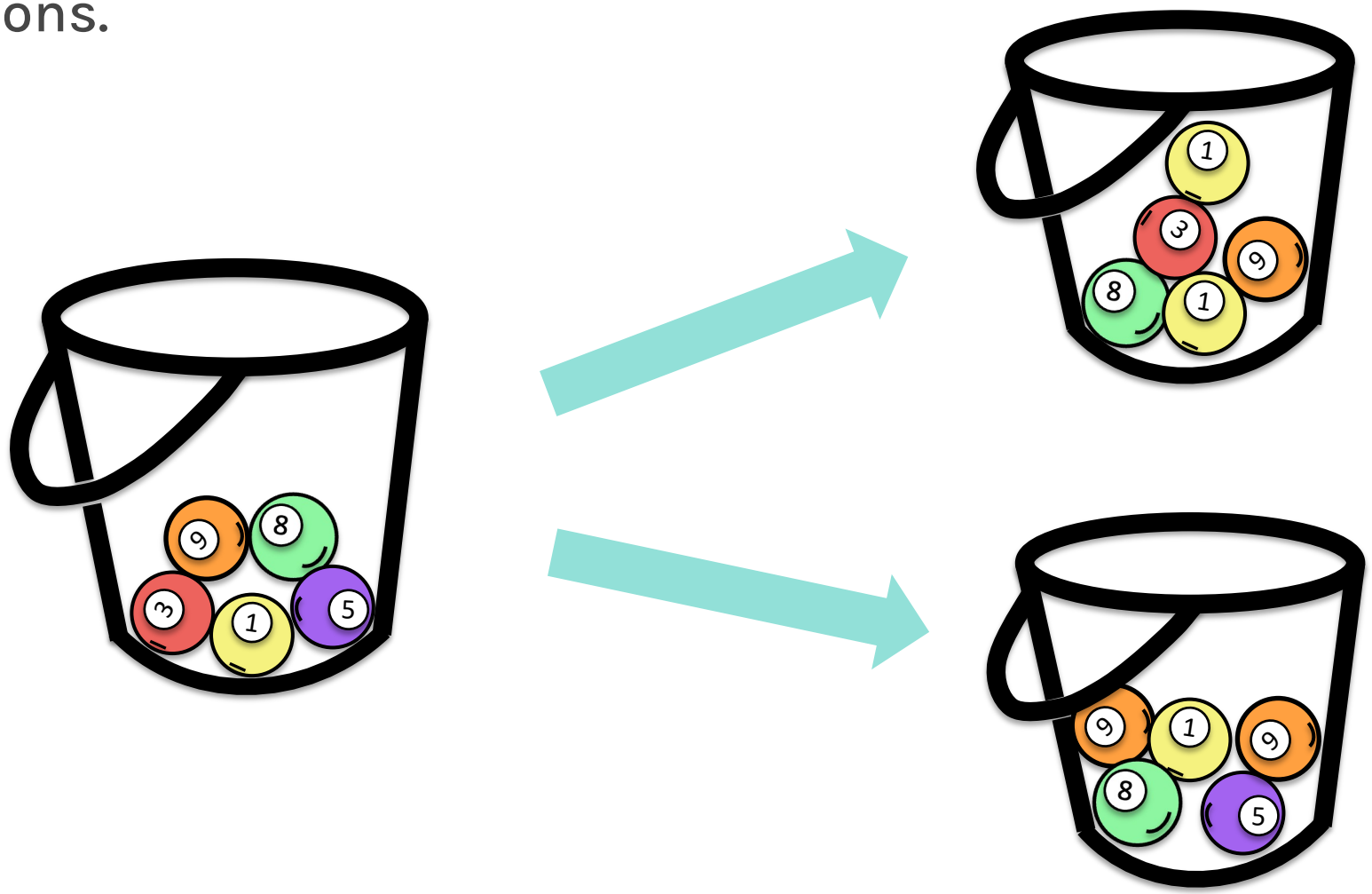
We continue until the “other” bucket has **the same number of balls** as the original one.



This new bucket represents a new parallel universe

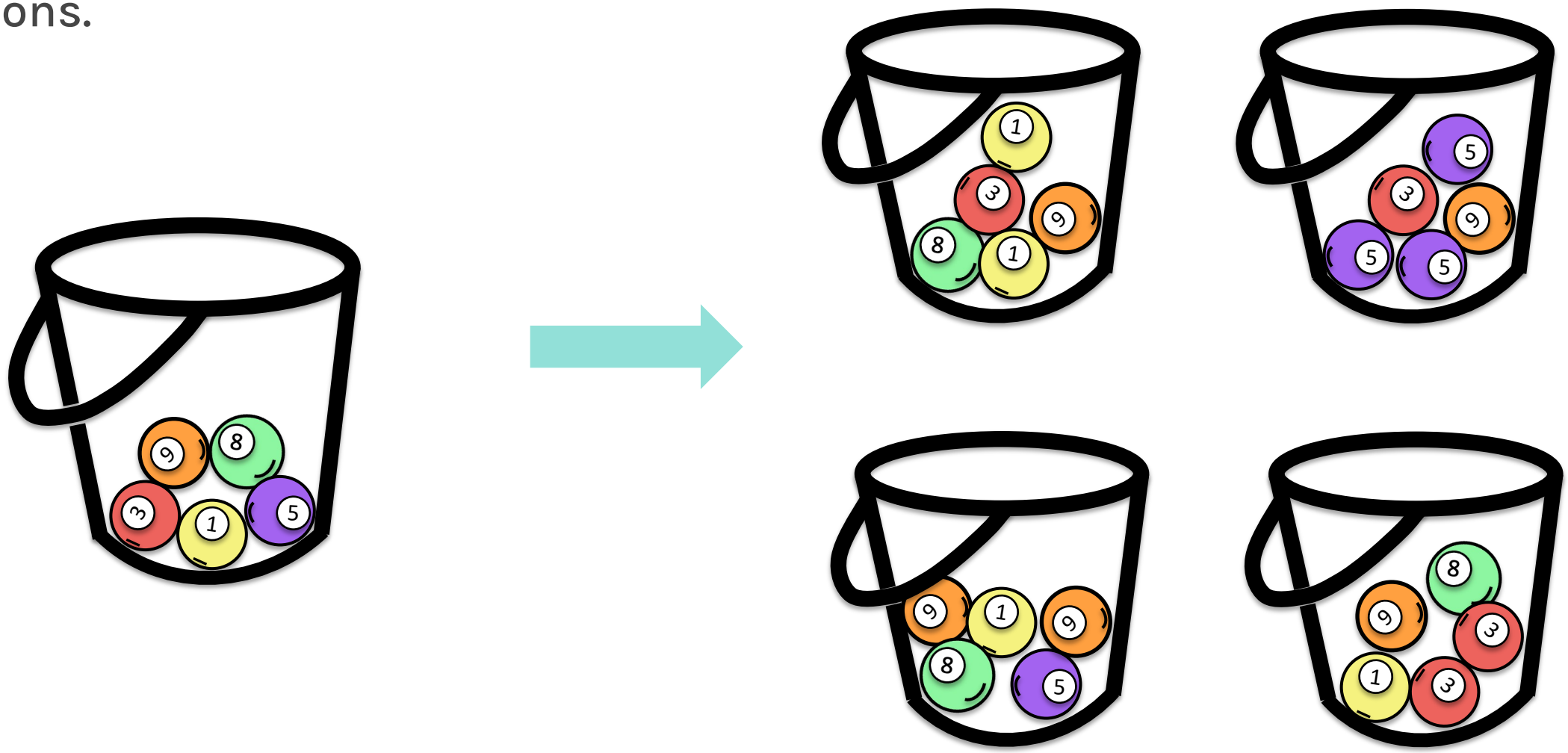
Bootstrap

We repeat the same process and acquire another set of bootstrapped observations.



Bootstrap

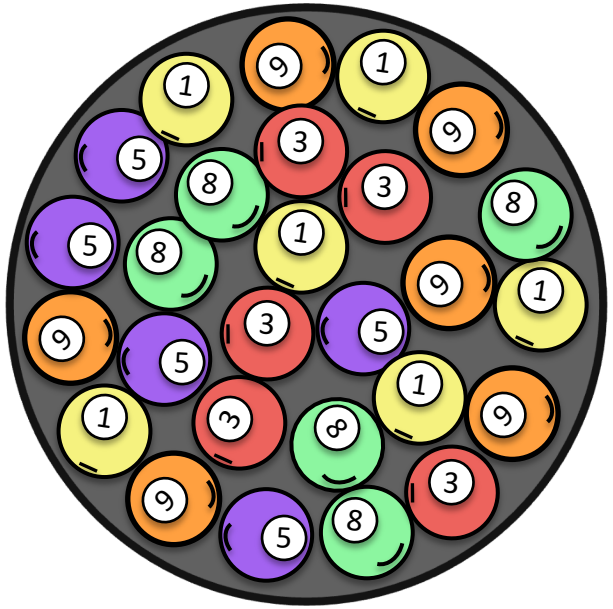
We repeat the same process and acquire another set of bootstrapped observations.



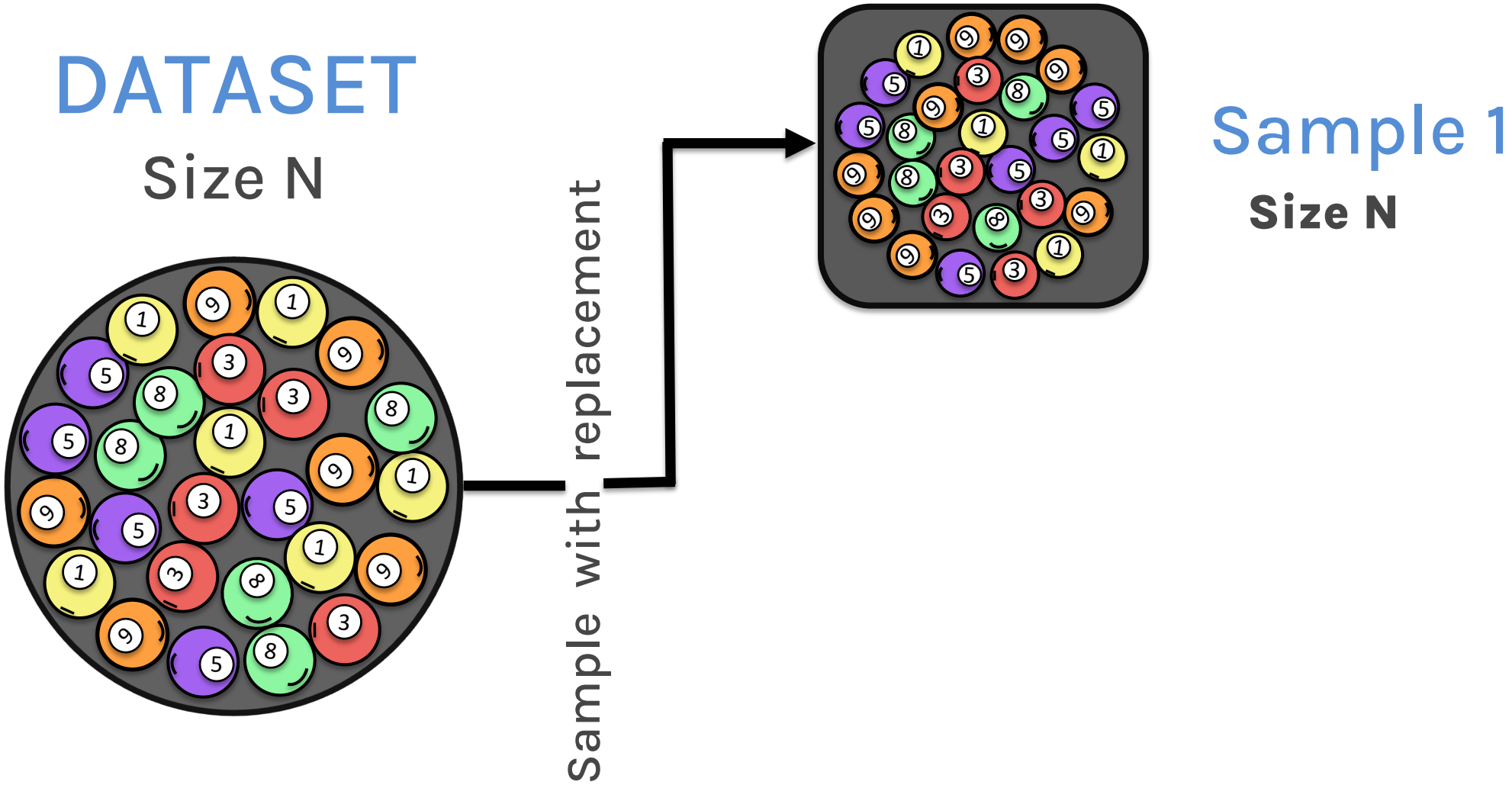
Bootstrap

DATASET

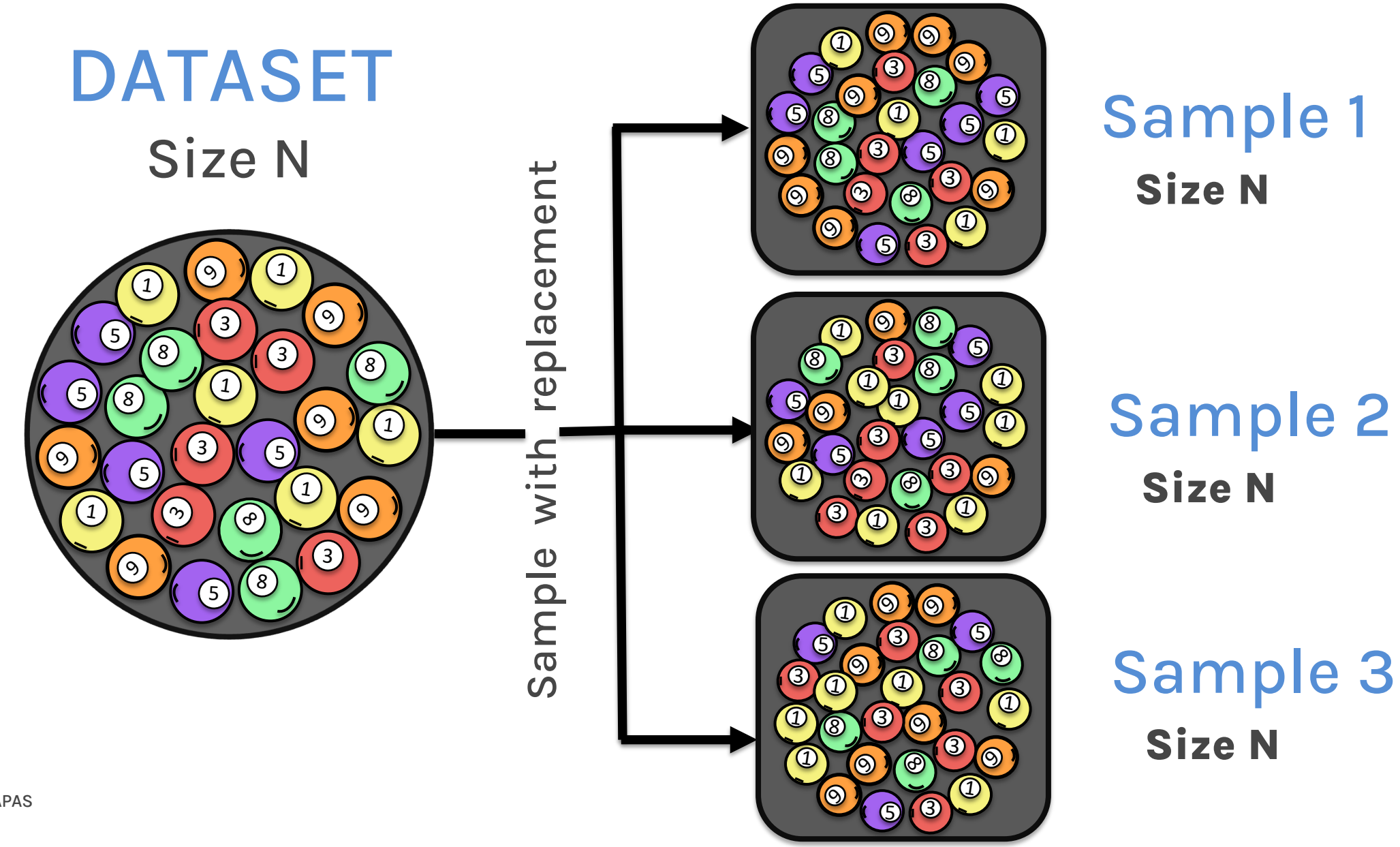
Size N



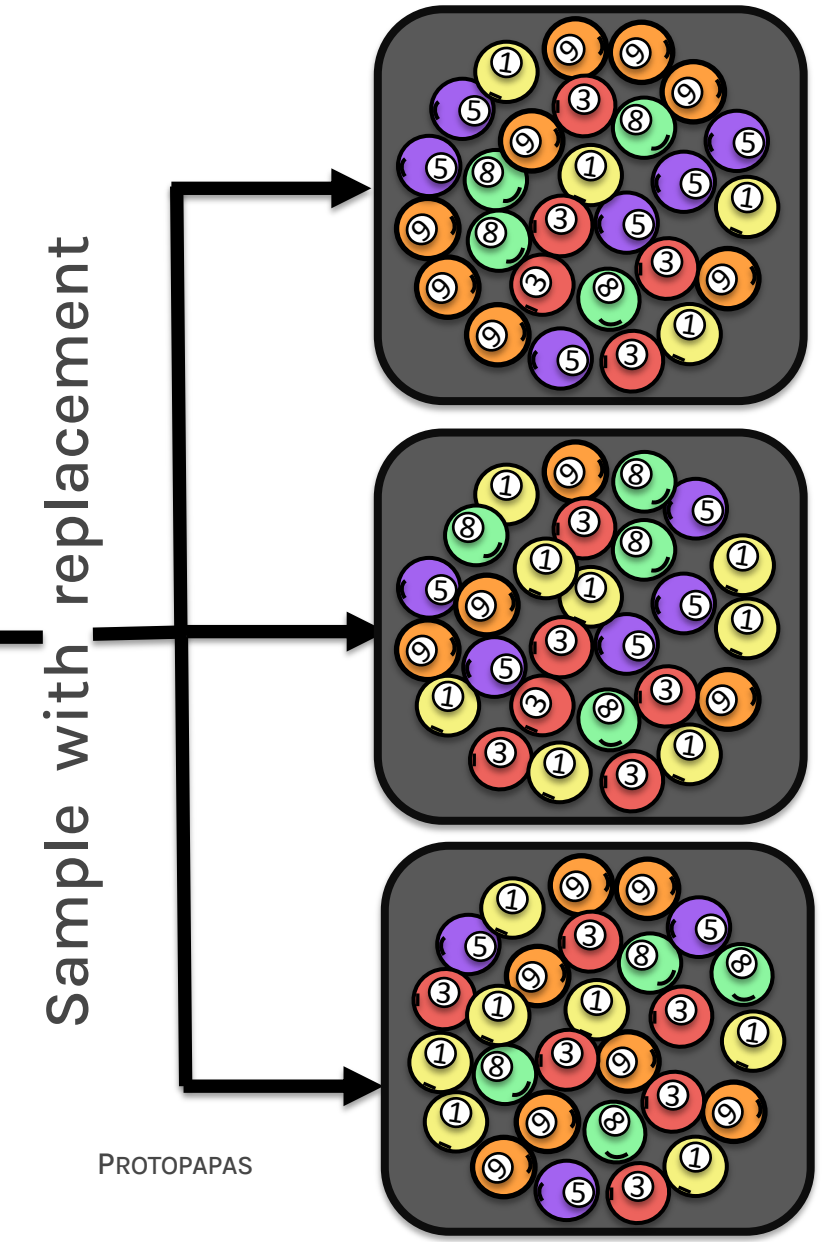
Bootstrap



Bootstrap


DATASET
Size N
Sample with replacement
Sample 1
Size N
Sample 2
Size N
Sample 3
Size N

Bootstrap



Sample 1

Size N

Train →

Model 1: $\hat{y} = \hat{\beta}_0^{(1)} + \hat{\beta}_1^{(1)} x$

Sample 2

Size N

Train →

Model 2: $\hat{y} = \hat{\beta}_0^{(2)} + \hat{\beta}_1^{(2)} x$

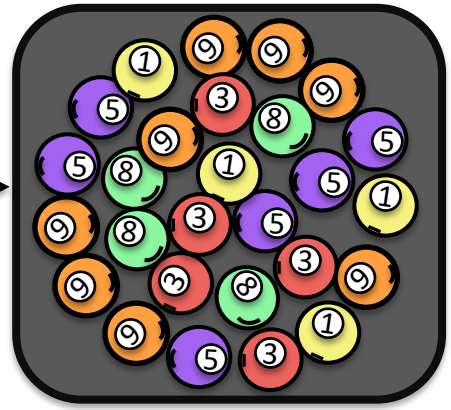
Sample 3

Size N

Train →

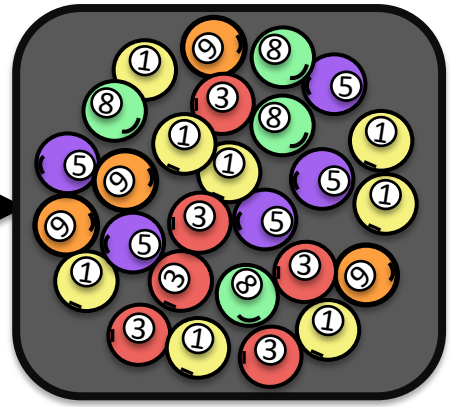
Model s: $\hat{y} = \hat{\beta}_0^{(s)} + \hat{\beta}_1^{(s)} x$

Bootstrap



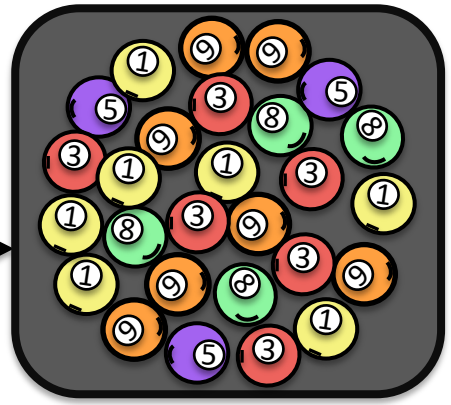
Sample 1 Train → Model 1: $\hat{y} = \hat{\beta}_0^{(1)} + \hat{\beta}_1^{(1)} x$

Size N



Sample 2 Train → Model 2: $\hat{y} = \hat{\beta}_0^{(2)} + \hat{\beta}_1^{(2)} x$

Size N



Sample 3 Train → Model s: $\hat{y} = \hat{\beta}_0^{(s)} + \hat{\beta}_1^{(s)} x$

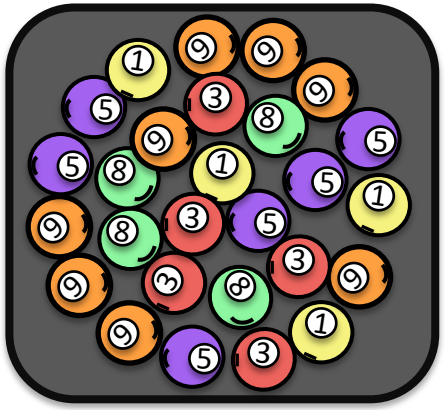
Size N

Combine models

$$\mu_{\hat{\beta}} = \frac{1}{s}\sum_{i=1}^{s} \hat{\beta}^{(i)}$$

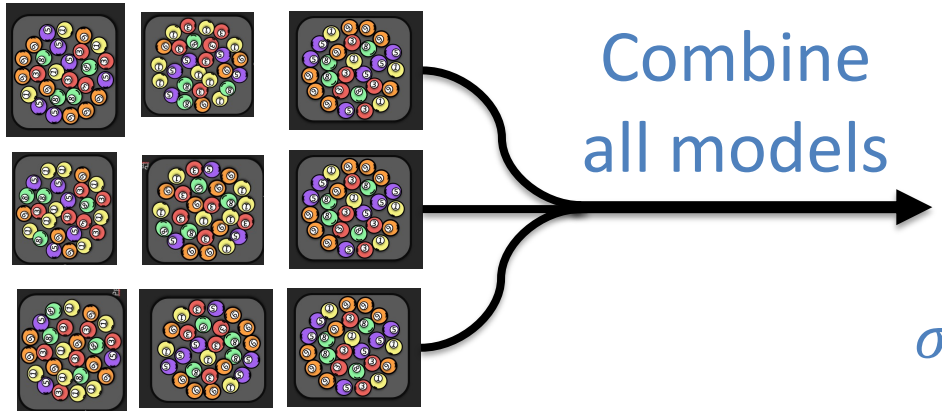
$$\sigma_{\hat{\beta}} = \sqrt{\frac{1}{s-1}\sum_{i=1}^{s} \left(\hat{\beta}^{(i)} - \mu_{\hat{\beta}}\right)^2}$$

In summary, for each “Parallel Universe”…



Train

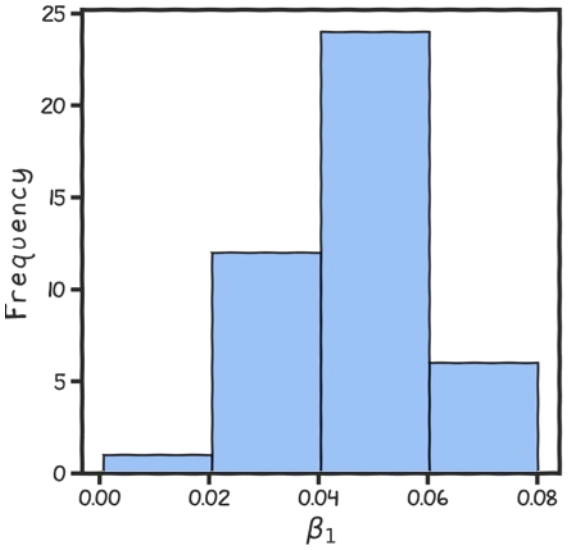
Model i: $\hat{y} = \hat{\beta}_0^{(i)} + \hat{\beta}_1^{(i)} x$



Combine all models

$$\mu_{\hat{\beta}} = \frac{1}{s}\sum_{i=1}^{s} \hat{\beta}^{(i)}$$

$$\sigma_{\hat{\beta}} = \sqrt{\frac{1}{s-1}\sum_{i=1}^{s}\left(\hat{\beta}^{(i)} - \mu_{\hat{\beta}}\right)^2}$$



s models

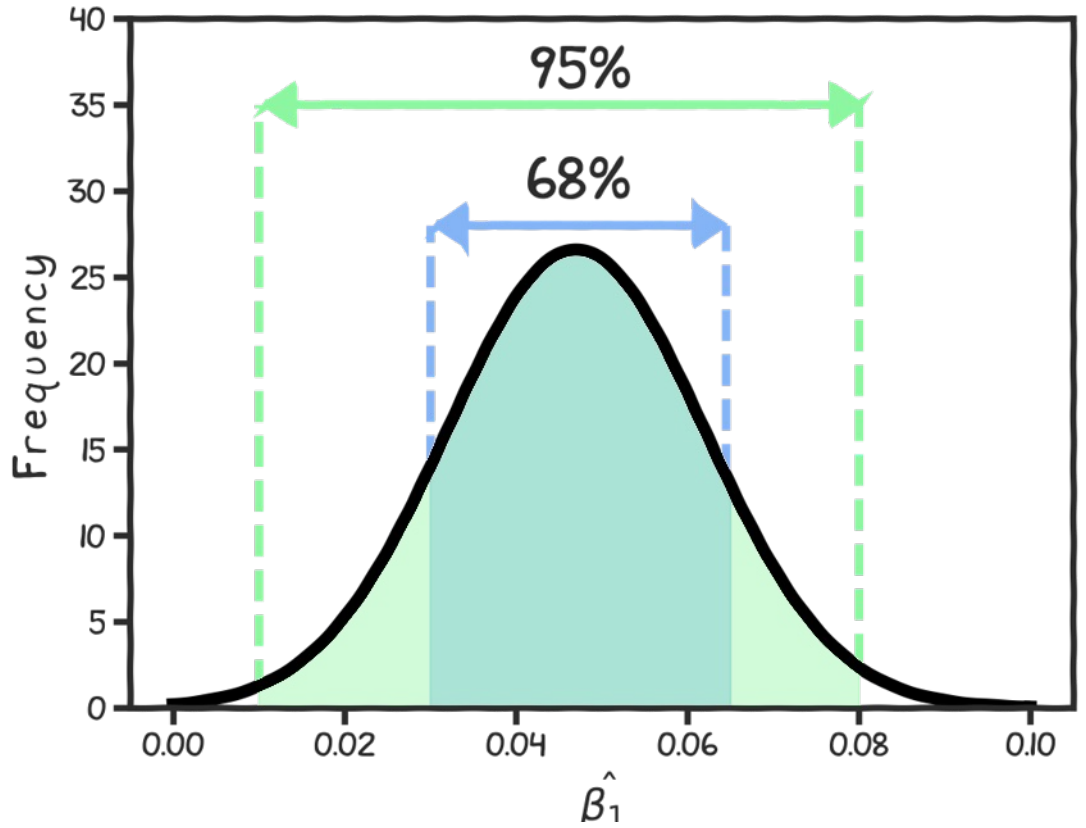
Bootstrapping for Estimating Sampling Error

Definition

Bootstrapping is the practice of estimating properties of an estimator by measuring those properties by sampling from the observed data.

For example, we can compute $\hat{\beta}_0$ and $\hat{\beta}_1$ multiple times by randomly sampling from our data set. We then use the variance of our multiple estimates to approximate the true variance of $\hat{\beta}_0$ and $\hat{\beta}_1$.

Confidence intervals for the predictor estimates (cont.)



The samples of $\beta's$ allow us to estimate the 95% confidence interval, which is the range of values that would contain the **true value** of $\hat{\beta}_1$ with 95% probability.

How do we calculate confidence intervals?

Confidence intervals for the predictor estimates (cont.)

Let's look at an example of how to construct a confidence interval for $\hat{\beta}_1$ using our bootstrap samples.

These numbers represent bootstrap estimates of $\hat{\beta}_1$ obtained by repeatedly resampling our dataset.

$$[13.75, 15.21, \ 13.65, 13.58, 12.93, 14.23, 12.81, \ 11.50, 13.09, 12.26, \ldots]$$



Step #1: Sort the bootstrap estimates of $\hat{\beta}_1$ from lowest to highest.

Confidence intervals for the predictor estimates (cont.)

[13.75, 15.21, 13.65, 13.58, 12.93, 14.23, 12.81, 11.50, 13.09, 12.26, ...]



Step #1: Sort the bootstrap estimates of $\hat{\beta}_1$ from lowest to highest.

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58, 13.65, 13.75, 14.23, 15.21, ...]



Step #2: Find the lower confidence range using np.percentile()

Confidence intervals for the predictor estimates (cont.)

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58,13.65,13.75, 14.23, 15.21,, …]



Step #2: Find the lower confidence range using np.percentile()

`np.percentile( [11.50, 12.26,12.81,12.93,13.09,13.58, 13.65, 13.75,14.23, 15.21,, … ], 2.5)` **= 12.80**

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58,13.65,13.75, 14.23, 15.21,, …]

2.5% of data are on the left of this value

Confidence intervals for the predictor estimates (cont.)

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58,13.65,13.75, 14.23, 15.21,, …]



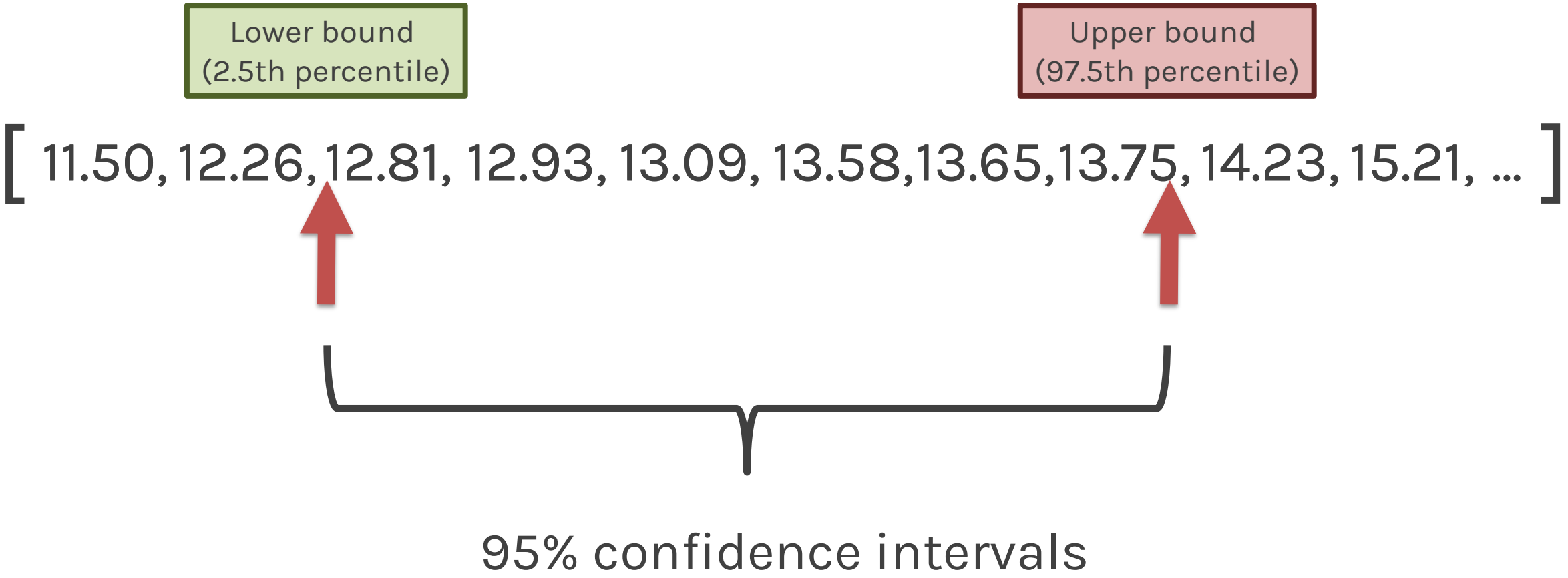
Step #3: Find the upper confidence range again using np.percentile()

`np.percentile( [11.50, 12.26,12.81,12.93,13.09,13.58, 13.65, 13.75,14.23, 15.21,, … ] ,97.5)` **= 13.71**

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58,13.65,13.75, 14.23, 15.21, …]

2.5% of data are on the right of this value

Confidence intervals for the predictor estimates (cont.)



Confidence intervals for the predictor estimates (cont.)

Lower bound
(2.5th percentile)

[11.50, 12.26, 12.81, 12.93, 13.09, 13.58, 13.65, 13.7 ... 6.21, ...]



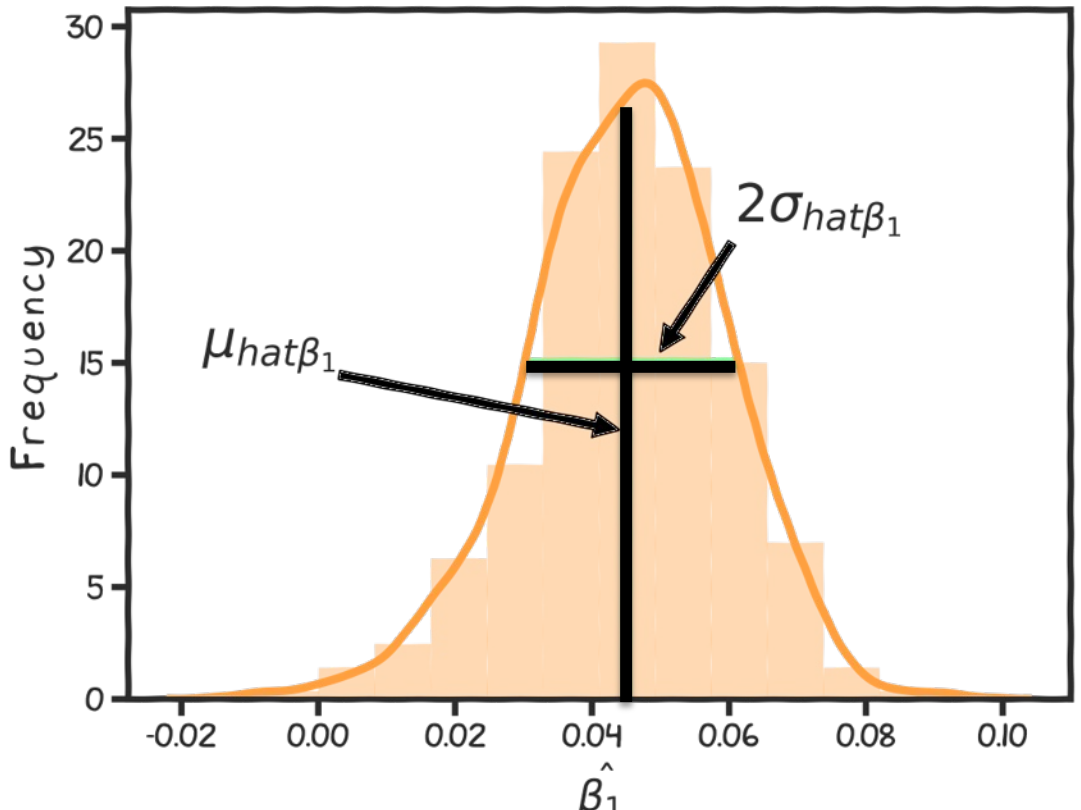
Confidence intervals provide bounds,
but the precision of $\hat{\beta}$ can also be **summarized** by its standard deviation, σ,
which we call the **standard error**.

95% confidence intervals

Confidence intervals for the predictors estimates (cont)

In other words we empirically estimate the standard deviations $\hat{\sigma}_{\hat{\beta}}$ which are called the **standard errors,** $SE_{\hat{\beta}_0}, SE_{\hat{\beta}_1}$ through bootstrapping.

Using these, one can calculate the 95% confidence intervals as approximately $\hat{\beta} \pm 2SE_{\beta}$.



We are making an assumption here. What is it?

Standard Errors based on probability theory

Alternatively: If we assume normality, then:

And if we know the variance σ_ϵ^2 of the noise ϵ, we can compute $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$ analytically using the formulae below (no need to bootstrap):

$$SE_{\hat{\beta}_0} = \sigma_\epsilon \sqrt{\frac{1}{n} + \frac{\mu_x^2}{\sum_i (x_i - \mu_x)^2}}$$

$$SE_{\hat{\beta}_1} = \frac{\sigma_\epsilon}{\sqrt{\sum_i (x_i - \mu_x)^2}}$$

$$\boldsymbol{CI_{\hat{\beta}}(95\%) = [\hat{\beta} - 2SE_{\hat{\beta}}, \hat{\beta} + 2SE_{\hat{\beta}}]}$$

Where n is the number of observations.

μ_x is the mean value of the predictor.

Standard Errors

In practice, we do not know the value of σ_ϵ since we do not know the exact distribution of the noise ϵ.

However, if we make the following assumptions:

- the errors $\epsilon_i = y_i - \hat{y}_i$ and $\epsilon_j = y_j - \hat{y}_j$ are uncorrelated, for $i \neq j$,
- each ϵ_i has a mean 0 and variance σ_ϵ^2,

then, we can empirically estimate σ_ϵ, from the data and our regression line:

$$\sigma_\epsilon = \sqrt{\frac{n \cdot MSE}{n-2}} = \sqrt{\sum \frac{\left(\hat{f}(x) - y_i\right)^2}{n-2}}$$

Remember: $y_i = f(x_i) + \epsilon_i \Longrightarrow \epsilon_i = y_i - f(x_i)$